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Paleomagnetism of the Upper Paleozoic of the Novaya Zemlya Archipelago

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Abstract—The rock magnetic and paleomagnetic results from the Upper Paleozoic sedimentary sequences composing the isles of the Novaya Zemlya Archipelago are presented. The recorded temperature dependences of the magnetic susceptibility, the magnetic hysteresis parameters, and the results of the first-order reversal curve (FORC) measurements suggest the presence of single-domain or pseudo-single-domain magnetite and hematite grains in the rocks. The Upper Paleozoic deposits overall are promising for unraveling the tectonic evolution of the Barents–Kara region. Together with the rock magnetic data, the positive fold and reversal tests testify to the primary origin of the indentified magnetization components. However, the interpretation of the paleomagnetic data should take into account the probable inclination shallowing. New substantiation is offered for the paleomagnetic poles for Early Devonian and Late Permian. For the first time, paleomagnetic constraints are obtained for the Late Carboniferous boundary. It is shown that the Early Cimmerian deformation stage within the Paikhoi–Novaya Zemlya region is associated with the sinistral strike slip displacement along the Baidaratskii suture during which the internal structure of the Southern Novaya Zemlya segment could undergo shear in addition to the nappe-thrust transformations. The Northern Novaya Zemlya segment, which is shifted northwest with respect to the Southern segment, was deformed in the thrusting mode with an overall clockwise rotation of this segment relative to the East European Craton.

Keywords: rock magnetism, paleomagnetism, Paleozoic, paleomagnetic pole, magnetotectonics, shear deformations, Barents–Kara shelf, Novaya Zemlya Archipelago, the Arctic

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INTRODUCTION

The tectonic evolution of the Barents–Kara region pertains to the most topical issues in the geological studies of the Arctic shelf. Despite long research and extensive geological and geophysical data obtained for this region, the present-day notions about the tectonics of this territory are still sketchy.

The existence of a large continental landmass—Hyperborean or Arctida—is the most substantiated scientific idea underlying the most of present-day tectonic reconstructions (Khain, 1979; Zonenshain and Natapov, 1987; Cocks and Torsvik, 2011; Vernikovsky et al., 2013a; Metelkin et al., 2015). This hypothesis was prompted by the comparability of the geological structure of the Paleozoic sections resting on a similar Precambrian metamorphic basement virtually across the entire self margin of the Arctic Ocean. According to the traditional view, Arctida was formed by the structures of the Central and Eastern Arctic—the Kara, New Siberian-Chukchi-Alaska, and Greenland-Ellesmere continental blocks, whose overall configuration was close to their present relative positions (Zonenshain and Natapov, 1987). Following this idea, it was suggested that Arctida also included Barentsia. Barentsia's northeastern part incorporating the islands of the Svalbard, Franz Joseph Land, and Novaya Zemlya archipelagoes is considered as the Grenville age Svalbard microplate, whereas the southeastern Timan–Pechora area with the adjacent Barents shelf is understood as a fragment of the Cadomian age orogen which stitches up the Svalbard plate with Baltica (East European craton) (Kuznetsov, 2008). However, the internal structure of the Arctic paleocontinent, the time of its assembly and breakup, the place occupied by the Svalbard plate in the general structure, and many other questions of the Pre-Mesozoic tectonics remain highly controversial.

As a rule, the notions concerning the tectonic history of Arctida and its constituents rely on the geological, petrological–geochemical, and, partly, geochronological data. Until recently, paleomagnetic determinations which contain the quantitative characteristics of the spatial location and drift kinematics of the continental blocks were almost absent. This was due to the

inaccessibility of Arctic islands and insufficiently extensive occurrence of the suitable rock complexes for the analysis. Today, the development of high-resolution instruments for paleomagnetic studies and the availability of the delivery means and survival facilities has started to gradually fill this gap. Inter alia, a series of the paleomagnetic determinations is obtained for the structures of the Central and Eastern Arctic (Metelkin et al., 2005; 2016; Vernikovsky et al., 2013b; Proskurnin et al., 2014; Zhdanova et al., 2016; Chernova et al., 2017a; 2017b). Due to this, it has become for the first time possible to quantitatively substantiate and develop the idea of Arctida (Vernikovsky et al., 2013a; Metelkin et al., 2015) and to outline the main directions and objects for study in the nearest future. The Paleozoic deposits of the Novaya Zemlya archipelago are an important element in this research. This is due to the completeness of the geological record for the Paleozoic interval of the geological history, the relatively easy accessibility of the sections, high-quality paleontological validation, and, besides, a number of the tectonic reasons. Here, in the southern part of the Pai-Khoi–Novaya Zemlya area, there is the Neoproterozoic suture which marks the edge of the Svalbard microplate (the Baidaratskii fault in Fig. 1), whereas the Paleozoic section simultaneously provides an insight into the history of Arctida, the Uralian Ocean, and the continental margin of Baltica. Besides, the position of the archipelago in the immediate vicinity of the junction with the Kara microplate is critically important for correlating the reconstructed events in the Central and Eastern Arctic to their counterparts in the Barents Sea region.

Hence, the Novaya Zemlya Archipelago is a key object for identifying the relationship between structural elements and reconstructing the tectonic history of the Barents–Kara sector of the Arctic Ocean in the Paleozoic, whereas the paleomagnetic investigation of the complexes composing it is a priority task for substantiating the geological history of Arctida and the global paleotectonic reconstructions.

The first paleomagnetic studies on the territory of the Novaya Zemlya islands were conducted in the 1980s under the guidance of A.N. Khramov. Until recently, the determinations obtained in these works (which will be discussed below) have been practically the only results for the entire Russian part of the Arctic. An important role in elucidating the geological history of the Arctic and, primarily, making the global magnetotectonic reconstructions was played by the catalogs of the World Data Center B published in the 1971–1989 under the editorship of Khramov. These catalogs summarized the data on the paleomagnetic directions and paleomagnetic poles for the territory of the former Soviet Union. These catalogs have long been the only source for creating the regional paleotectonic schemes and reconstructions. For some regions of the former Soviet Union, they remain relevant even now.

This paper continues the paleomagnetic studies of the Novaya Zemlya Archipelago which were started at the end of the 20th century; it addresses the study of the Upper Paleozoic of the Northern and Southern islands of this archipelago.

BRIEF GEOLOGICAL CHARACTERISTIC OF THE STUDY REGION

The structures of the Novaya Zemlya Archipelago compose the main part of the Pai-Khoi–Novaya Zemlya fold-and-thrust region which separates the recent sedimentary basins of the Barents and Kara shelves, on the northern continuation of the Polar Urals. The structural pattern of the region is determined by the arcuate configuration of the complex system of the folded and fault structures skirting the South Kara Basin (Fig. 1). Two blocks are distinguished in the structure of this territory. The southern Neoproterozoic block (with the Cadomian consolidation) is associated with the Bol'shezemel'skaya part of the Timan–Pechora plate (Kuznetsov et al., 2005; Sharov et al., 2005), and the northern Mesoproterozoic block (with the Grenville consolidation) is correlated to the Svalbard Plate (Korago et al., 1992; Korago and Timofeeva, 2005). The Baidaratskaya suture zone is the boundary between these blocks (Fig. 1). The fold nappes are formed by the Paleozoic sedimentary complexes (dislocated during the Early Cimmerian (pre-Jurassic) orogeny) overlying this heterogeneous base (Korago et al., 1992). The deposition of these complexes is associated with the continental shelf conditions of Baltica in the margin of the Uralian paleoocean. The early Paleozoic is typically ubiquitously dominated by the carbonate flysch sedimentation (Sobolev et al., 1985; *Gosudarstvennaya*..., 2014). In the Middle Devonian, the paleobasin was divided with the formation of two facial zones. A deep-water trough was formed in the eastern and central part of the paleobasin, where bathial carbonate-siliceous-clayey sediments were deposited. A shallow carbonate shelf area was preserved in the southwestern and western parts of the region. The inception of the trough was accompanied by basite magmatism with the petrochemical features of the present-day oceanic tholeiites, which was synchronous with the rifting processes in the Timan–Pechora region, in Pai-Khoi, and in the Barents Sea (Basov et al., 2009).

During the second half of the Carboniferous and in the Permian, the north of the Novaya Zemlya accommodated the formation of the thin coastal-marine terrigenous layers. The rest of the territory to the end of the Carboniferous has become an area of ubiquitous deep-sea sedimentation (Basov et al., 2009). In the Late Permian, the deep-sea trough began to be filled with the clastic material abundantly supplied from the Uralian orogen. The development of the trough was completed toward the middle of the Triassic and the entire region experienced another episode of strong

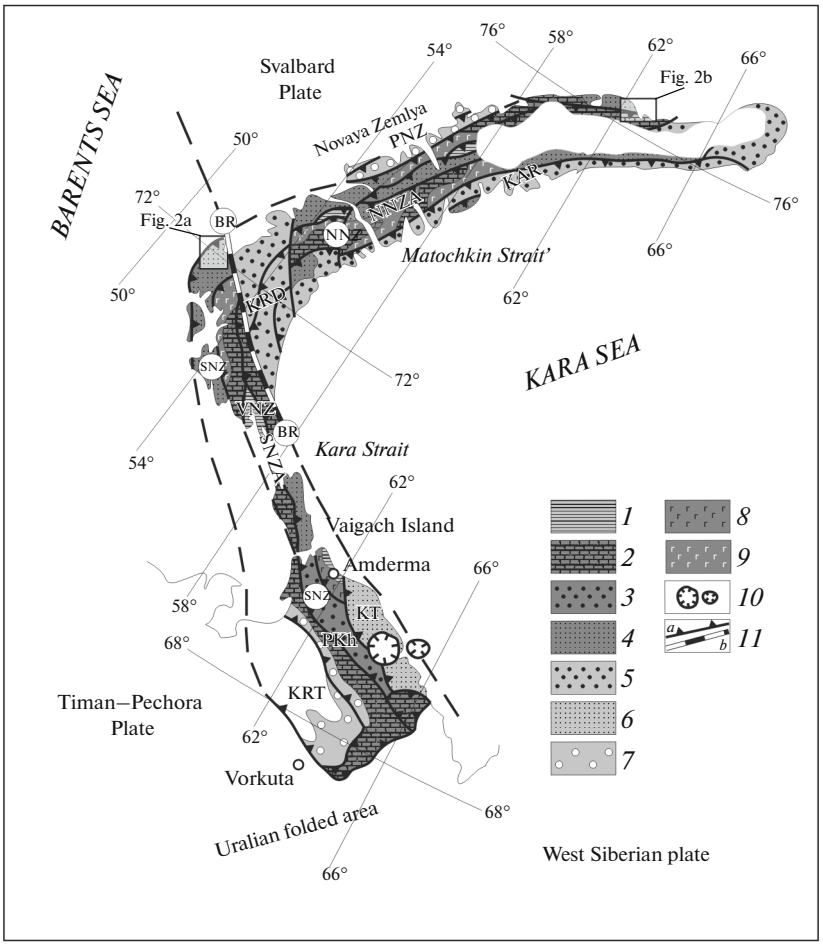


Fig. 1. Geodynamics map for Pai-Khoi–Novaya Zemlya fold-and-thrust region: (*1*) uplifts of Precambrian metamorphic basement rocks; (*2–4*) Early Paleozoic passive continental margin deposits for Southern Novaya Zemlya segment (SNZ) and Vendian–Early Paleozoic age for Northern Novaya Zemlya segment (NNZ): (*2*) carbonate and carbonate-terrigenous shelf deposits, (*3*) flysch-like and turbidite rocks of continental slope and continental rise; (*4*) terrigenous shelf and coastal-marine; (*5–6*) Late Paleozoic–Early Mesozoic passive continental margin deposits; (*5*) flysch-like and turbidite rocks of continental slope and continental rise; (*6*) terrigenous shelf and coastal-marine rocks; (*7*) Triassic molasse; (*8–9*) occurrences of Paleozoic intraplate magmatism: (*8*) related to rifting; (*9*) related to trap magmatism; (*10*) astroblemes; (*11*) (*a*) large thrusts and (*b*) Baidaratskii (BR) collisional suture; dashed line shows their presumed continuation. Letter denotations: NNZA, Northern Novaya Zemlya antiform; SNZA, Southern Novaya Zemlya antiform, including Vaigach–Novaya Zemlya (VNZ) and Pai-Khoi (PKh) highs; troughs: KT, Kara Trough and KRT, Korotaikha Trough; and depressions: KAR, Kara Depression; KRD, Karamakuly Depression; and PNZ, Pre-Novaya Zemlya Depression.

deformations which created its present-day structural fabric (Fig. 1).

The structure of the Southern Novaya Zemlya segment of the region is determined by a series of northwesterly trending tectonic nappes. Within the Pai-Khoi and Vaigach–Southern Novaya Zemlya uplifts forming a single Southern Novaya Zemlya antiform, the inliers of the Cadomian folded basement are observed. They are sharply discordant, with stratigraphic inconsistency in the Cambrian section, and are overlain by the Early Paleozoic sedimentary complex. The Late Paleozoic part of the section gravitates to the periphery of the antiform, whereas the youngest Permian–Triassic sequences fill the Korotaikha trough bordering the antiform in the west (Fig. 1).

The Northern Novaya Zemlya segment includes the continuous Late Precambrian–Early Mesozoic sedimentary complex resting on the Grenville crystalline basement of the Svalbard Plate. Its northeast trending fold–nappe structures frontally abut the Baidaratskii fault, which was reactivated at the Early Cimmerian deformation stage. The central part of the segment is also most elevated and forms the Northern Novaya Zemlya antiform. Correspondingly, the Late Paleozoic formations predominantly occur along the coastal areas and within the Karamakuly depression neighboring the Baidaratskaya fault zone (Fig. 1).

In the expedition of 2014 which was organized by the Karpinsky Russian Geological Research Institute (VSEGEI) with support from OOO RN-Shel'f-Arktika,

the Late Paleozoic sections were studied in two regions of the Novaya Zemlya Archipelago: in the Gusinaya Zemlya Peninsula in the Southern Novaya Zemlya part and the region of the Russkaya Gavan Bay within the Northern Novaya Zemlya segment (Fig. 1).

At the northern tip of the Gusinaya Zemlya Peninsula, the bedrock outcrops expose the Middle–Upper Permian and Triassic sediments which, together with the other formations, compose the structure of the large Northern Gusinskaya synclinal fold with the meridionally striking axis and north plunging hinge (Fig. 2a). We have studied the coastal outcrops of two formations: the Shadrovskaya and Sedoyakhskaya ones (the latter is also referred to as Admiralteiskaya in the other stratigraphioc schemes) in the western limb of the fold.

The Shadrovskaya Formation is represented by the coastal-marine deposits: conglomerates, sands, and silts with coal interbeds and flora detritus in the top portions of the section (*Putevoditel'...*, 2014). Crossbedded bar sandstone members are characteristic in it. Besides large flora remains, the rocks of this formation contain a rich variety of the foraminifera, *Neogeoceras* sp. ammonoidea, *Spitzbergenia alferovi* (Milor.), *Cancrinella* aff. *spitzbergenia* (Gobb.), and *Megousia weyprechti* (Toula) brachiopods, and pelecypods of the end of Middle Permian–Upper Permian age (Pavlov et al., 1981; 1999). The formation's thickness reaches 600–850 m. Two close outcrops of massive dark gray sandstones were sampled for the paleomagnetic studies (sampling points 14NZ01 and 14NZ02).

The 1100-m thick Sedoyakhskaya Formation conformably overlays the Shadrovskaya stratum and is composed of coastal–marine and continental, predominantly red, rhythmic sediments. Sandstones with interbeds and lenses of conglomerates are present at the base of the rhythms. Above them there are siltstone-mudstone members with subordinate sandstones, silty sandstones, basic tuffites, and tuffs. Here, the tufaceous rocks gravitate to the bottom part of the section. The rhythms typically end with dark gray and cherry red mudstones (Cherkesov and Makarov, 1982). Petrified remains are rare and represented by pelecypoda, vertebrate fossils, vegetable detritus, and an Early Triassic spore-pollen complex characterizing the uppermost Tatarian deposits of the Permian and the lowermost Triassic rocks (Cherkesov and Kasatkina, 1982). According to the set of the geological and magnetostratigraphic data, the Permian/Triassic boundary is located 100 m above the base of the formation (Gurevich and Slautsitais, 1984). We have studied thin interbeds of gray massive sandstones intercalated with finely tabular cherry-red siltstones near this boundary (sampling point 14NZ03). The complex of the presented stratigraphic data constrains the age of the rocks sampled for the paleomagnetic analysis to the interval from 260 to 250 Ma ago.

The works in the northern part of the Northern island of Novaya Zemlya were conducted at three sites (Fig. 2b). In the Cape Nassau region, in the Vykhodnaya riverbed, paleomagnetic samples were taken from two outcrops of the Barents Formation. Based on the macrofauna (foraminifera, bryozoans, brachiopods, ammonoidea) contained in the lower subformation, which is composed of limestones, sandstones, siltstones, and mudstones, these layers refer to the Middle–Upper Carboniferous (Sobolev et al., 1985). According to the biostratigraphic studies, the upper subformation which is composed of sandstones, sandy limes, and silts is dated to the Early Permian (Sobolev et al., 1985). The total thickness of the formation is over 1000 m. Both sampling points correspond to the upper horizons of the lower subformation and are located near the supposed Carboniferous/Permian boundary, which constrains their age by the interval from 310 to 300 Ma ago. Dark gray crossbedded sandstones are sampled at the 14NZ07 sampling point. From the fine alternating red and greenish gray parallel-bedding sands at the 14NZ08 sampling point, the samples are predominantly taken from red interbeds.

The Paleozoic sedimentary section east of the Russkaya Gavan Bay, in the region of Cape Consolation (Cape Utesheniya) and Chaev's Bay includes a complex of the Silurian, Devonian, and Lower Carboniferous terrigenous-carbonate rocks (Fig. 2b). Our study was mainly focused on the Lower Devonian rocks of the Ermolaevskaya Formation and the Middle Devonian sands of the Konglomeratovaya Formation.

The Konglomeratovaya Formation was studied at Cape Konglomerat in the region of Voronina Bay (sampling point 14NZ05) (Fig. 2b). The formation is dominated by the polymictic conglomerates with a maximal thickness of ~300 m. Well-rounded pebble contains dolomites (up to 60%), limestones (up to 20%), sandstones and siltstones (up to 10%), and vein quartz (about 9%). The organic remains from pebble and boulders are represented by corals, brachiopods, and ostracods characteristic of the Lower and Upper Silurian, as well as the Lower and Middle Devonian of the Urals (Sobolev, 1984; Sobolev et al., 1985). In the region of Cape Konglomerat, in the base of the formation, there is a unit of gray medium- to thick-bedded fine-grain clayey limestones, including the organogenic ones and black mudstones with intercalations of silts, sands, and sandy limes with a thickness of about 50 m. Samples for paleomagnetic analysis were taken from the layer of greenish gray medium-grain sandstones in the transition zone of this unit to the conglomerates. The Konglomeratovaya formation is dated to the Givetian but its age can be somewhat younger—Frasnian (Korago et al., 1992; *Putevoditel'...*, 2014). Based on this we assume that the age of the studied rocks can be in the interval from 388 to 372 Ma.

The Ermolaevskaya Formation was studied on the southeastern coast of Lake Retovskoe at the foot of the

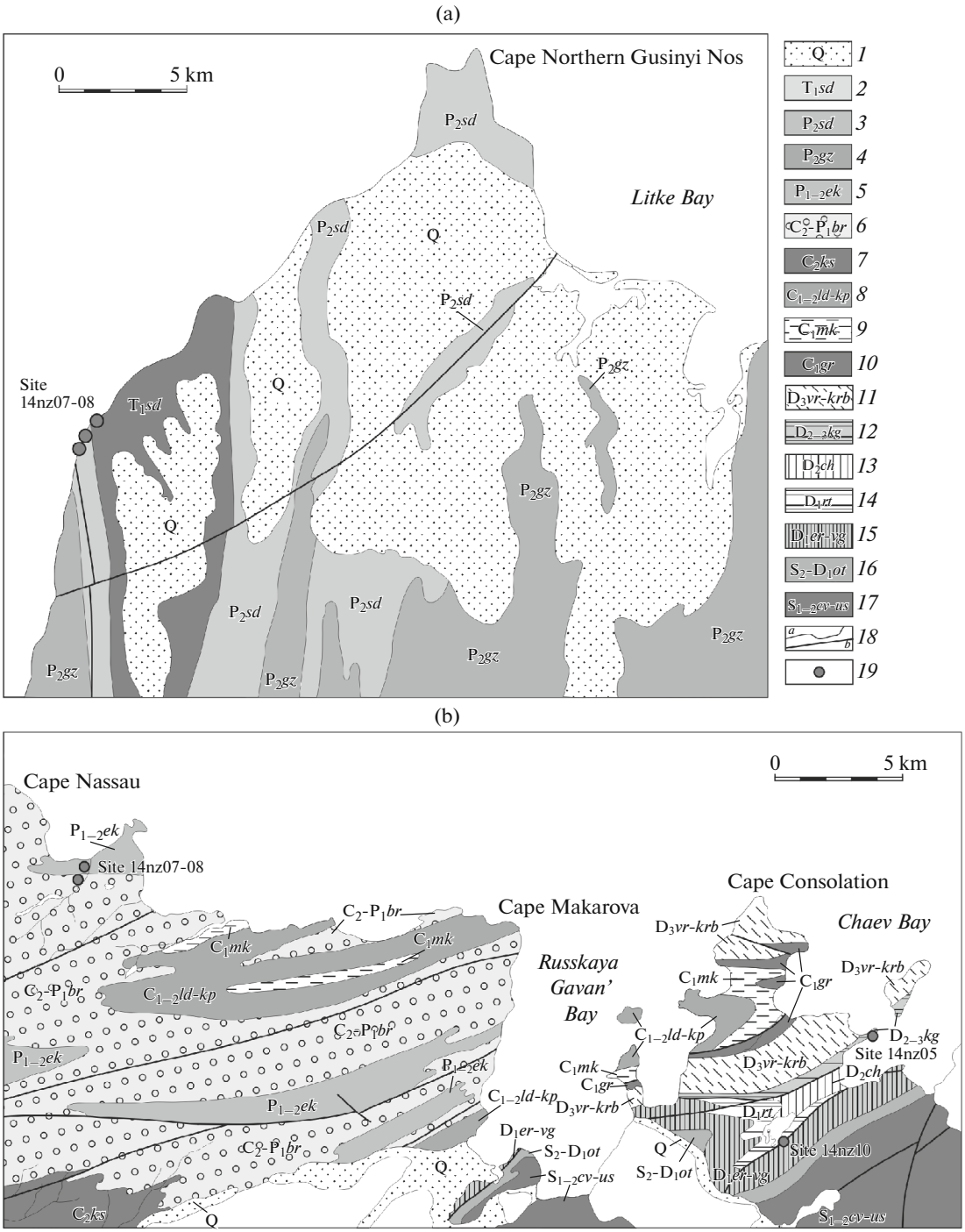


Fig. 2. Scheme of geological structure for studied areas of Novaya Zemlya Archipelago according to (*Gosudarstvennaya*..., 2013; 2014): (*1*) Quaternary pebblestones, sands, clays, and peat; (*2*) Lower Triassic Sedoyakhskaya Formation: sandstones and siltstones; (*3*) Upper Permian Gysinaya Zemlya Formation: mudstones, siltstones, sandstones; (*4*) Upper Permian Gusinaya Zemlya Formation: mudstones, siltstones, sandstones; (*5*) Lower and Upper Permian Eksovskaya Formation: mudstones, siltstones, sandstones; (*6*) Middle Carboniferous–Lower Permian Barentsevskaya Formation: sandstones, mudstones; (*7*) Middele Carboniferous Kseninskaya stratum: mudstones, sandstones; (*8*) unstratified Lower–Middle Ledyanaya Gavan and Kapriznинskaya formations: limestones, mudstones, siltstones; (*9*) Lower Carboniferous Goryakovskaya Formation: carbonate breccia, limestones, calcarenites; (*10*) Lower Carboniferous Goryakovskaya Formation: carbonate breccia, limestones, calcarenites; (*11*) unstratified Upper Devonian Voroninskaya and Karbasnikovskaya formations: limestones, mudstones, and dolomites; (*12*) Middle–Upper Devonian conglomerate formation: conglomerates and sandstones, limestones; (*13*) Middle Devonian Chukhnovskaya Formation: mudstones, siltstones, limestones; (*14*) Lower Devonian Retovskaya Formation: dolomites and limestones; (*15*) unstratified Lower Devonian Ermolaevskaya, Kan'onnaya, and Veselogorskaya formations: sandstones, siltstones, mudstones; (*16*) unstratified Upper Silurian–Lower Devonian Otkupshchikovskaya Formation: limestones, siltstones; (*17*) unstratified Lower–Upper Silurian Chaevskaya and Usachevskaya formations: siltstones, limestones, sandstones; (*18*) geological boundaries: (*a*), stratigraphic; (*b*), tectonic; (*19*) paleomagnetic sampling points.

Veselye mountains (point 14NZ10 in Fig. 2b). The formation is distinguished by the variegated coloration of the rocks and is represented by rhythmically alternating sandstones of different compositions, including the quartzite varieties, silts, and dolomites, with clearly predominant sandstones. The formation has a total thickness of ~600 m and is typically subdivided into three parts. We sampled the outcrop of the red fine-grained sandstones in the intermediate part of the formation. The age of the Ermolaevskaya rocks, which is only determined stratigraphically by its position in the section, is Lochkovian–Praghian (419–407 Ma) (*Putevoditel'...*, 2014).

RESULTS OF ROCK MAGNETIC STUDIES

Several samples from each sampling site were investigated for the composition and the domain state of the magnetic mineral grains presented in the rock. The magnetic hysteresis loops (Fig. 3a) and first-order reversal curves (FORC diagrams) were measured by a 3900 VSM vibration magnetometer (LakeShore Cryotronics, USA) in the geomagnetic laboratory of the Center for Earth Evolution and Dynamics (CEED) of the University of Oslo, Norway. The data were processed and analyzed using the FORCinet software (Harrison and Feinberg, 2008). The ratios of the main parameters (saturation magnetization M_s, saturation remanence M_{rs}, coercivity H_c, and coercivity of remanence H_{cr}) are illustrated in the Day plot (Day et al., 1977). FORC analysis provides more detailed information about the magnetic particle distribution in the rock by the coercivity and about the magnetic interaction. In particular, this method is capable of distinguishing a mixture of single-domain (SD) and multidomain (MD) particles from an ensemble of true pseudo single-domain (PSD) grains. The physical sense of the FORC-diagrams and their interpretation are thoroughly described in (Roberts, 2000; Roberts et al., 2014). According to the FORC diagram (Fig. 3b) and Day plot (Fig. 3c), the magnetic minerals in the studied rocks are in the SD or PSD state. The FORC diagrams for an ensemble of SD-particles have the shape of a symmetrical distribution of the contour along the H_c axis, without a vertical scatter along the H_u axis. On the contrary, the PSD particles have an asymmetric distribution with a contour diverging along the vertical axis (Roberts et al., 2014). The SD state is observed in most of the sandstone samples of the Barents Formation (sample 14nz095 in Fig. 3) and in the Lower Devonian sandstones of the Konglomeratovaya and Ermolaevskaya formations (sample 14nz108 in Fig. 3). The rocks of the Shadrovskaya and Sedoyakhskaya formations are dominated by PSD grains (sample 14nz031 in Fig. 3). Here, the PSD samples typically have higher H_c and H_{cr} (with a low H_{cr}/H_c ratio) which can be caused by the partial oxidation of the primary detrital magnetite in these rocks.

The experiments on studying the temperature dependences of magnetic susceptibility in an inert (argon) environment were carried out with the use of the MFK1-FA multifunctional kappabridge (AGICO, Czech Republic) in the Laboratory of geodynamics and paleomagnetism of the Trofimuk Institute of Petroleum Geology and Geophysics, Siberian Branch, Russian Academy of Sciences, Novosibirsk. In all the studied rocks, heating caused irreversible changes (Fig. 4). The cooling curves are located significantly higher than the heating curves; magnetic susceptibility increases after the experiment by one–two orders of magnitude compared to the initial value. This clearly testifies to the crystallization of a significant amount of the secondary magnetic phase which, as suggested by the Curie temperatures determined from the cooling curves, is magnetite. In separate cases (Fig. 4a), we might expect the crystallization of a new magnetic mineral with a lower Curie temperature, which can barely be diagnosed from these graphs. Iron for the newly formed magnetic phases can be supplied from iron sulfides, primarily pyrite, which was frequently observed in the older sedimentary rocks of the archipelago in the field studies.

The behavior of magnetic susceptibility during the first heating and cooling shows that the studied rocks can be conditionally subdivided into three types. In the *first type* of rocks, magnetic susceptibility monotonically decreases almost to 580°C after which it drops sharply and starting from 600°C again gradually declines to the end of heating to ~680°C (Fig. 4a). This suggests that the initial magnetic substance is represented by both magnetite and hematite. The presence of hematite which does not change under heating is supported by the repeated drop of the magnetic susceptibility under cooling in the high-temperature interval. The cause of the crystallization of the secondary magnetic phases is not quite clear because any significant rise of the graph $k(T)$ is absent. The described pattern of the changes in the magnetic susceptibility during the experiment is mainly observed for the sandstones of the Shadrovskaya and Sedoyakhskaya formations.

The second type of the $k(T)$ graphs has a smooth decrease in susceptibility under heating to ~400°C or slightly higher. Here, the initial magnetic susceptibility is noticeably lower (Fig. 4b). The further heating to ~560°C leads to the appreciable growth in magnetic susceptibility and only after it, the susceptibility sharply drops to zero at ~600°C. The cooling curve shows that after heating the sample only contains magnetite. The signs of the presence of the other magnetic minerals are neither revealed during heating nor cooling. This type of graphs is characteristic of the sandstones of the Barents Formation.

The third type, observed for sandstones of the Ermolaevskaya and Konglomeratovaya formations is distinguished by the somewhat more complicated

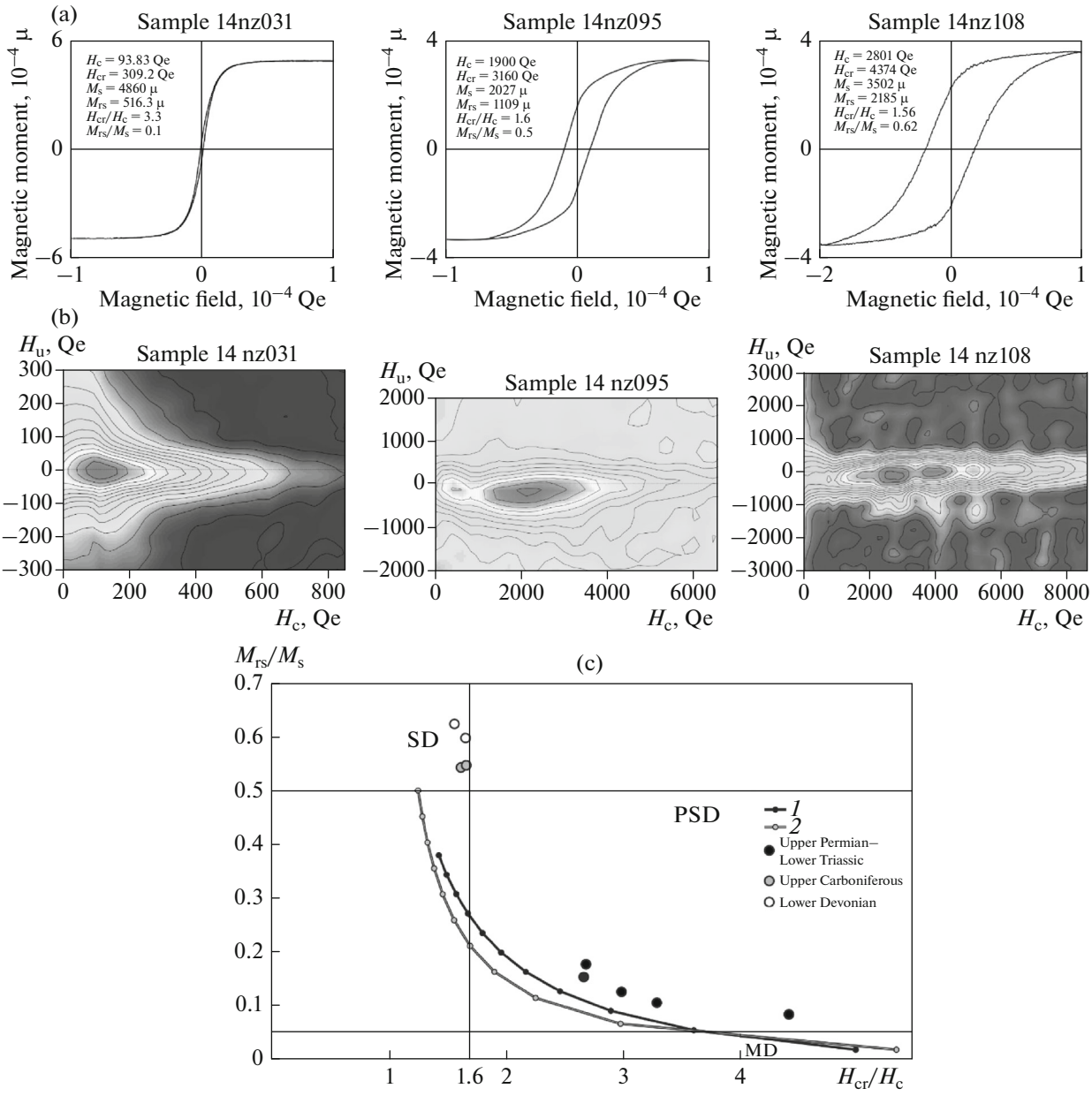


Fig. 3. (a) Magnetic hysteresis loops after elimination of paramagnetic component; (b) FORC diagrams; (c) Day plot for studied rocks. In Day plot, curves *1* and *2* are based on experimental data for (*1*) titanomagnetite and (*2*) magnetite as function of percentage of multidomain and single domain grains (Dunlop, 2002); SD, single-domain; PSD, pseudo single domain; MD, multidomain.

behavior of magnetic susceptibility during the experiment (Fig. 4c). Firstly, the crystallization of the new magnetic minerals starts even at 320–330°C, and the magnetic susceptibility grows in a stepwise manner. Secondly, the abrupt drop in susceptibility at heating to ~540°C is followed by the gradual decrease up to 660°C, which might testify to the presence of hematite. However, the coincidence of the cooling curve with the heating curve, which is characteristic of this mineral, is not observed. Susceptibility becomes different from zero only at heating to ~600°C after which it grows sharply and acquires values that are order of magnitude higher than the initial values (Fig. 4c). This peculiarity in the behavior of this rock magnetic parameter is most likely associated with the presence of oxidized magnetite grains in the stressed state, which causes heightening of its Curie temperatures under heating. Heating in an inert atmosphere results in the reduction of the oxidized grains to the initial magnetite, and the critical temperatures observed during the cooling correspond to the standard temperatures.

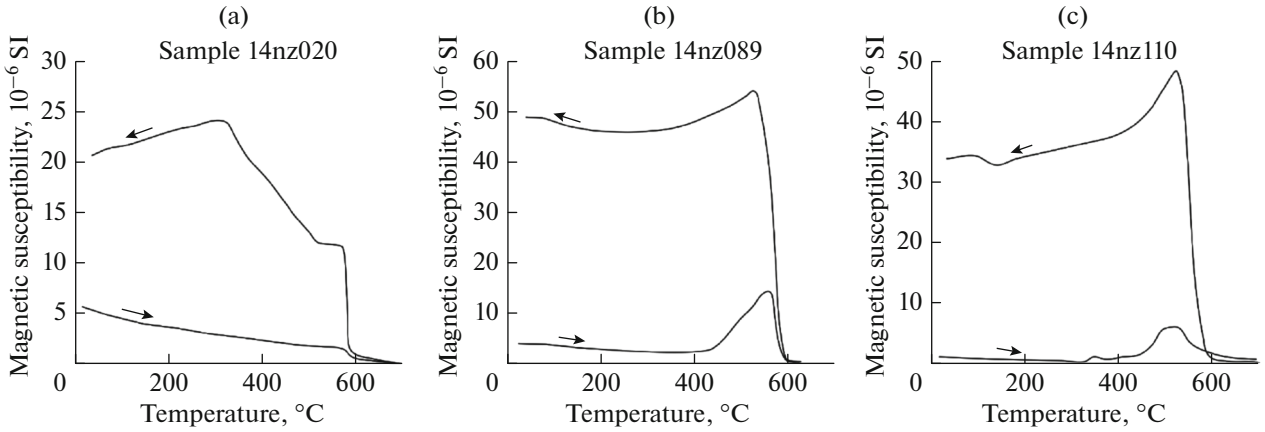


Fig. 4. Results of thermomagnetic analysis of studied rocks. Right and left arrows indicate heating and cooling, respectively.

Therefore, the results of our rock magnetic studies suggest that the main magnetic mineral in all the studied sandstones is magnetite which is in the hard SD state in the Devonian and Carboniferous deposits. The Late Permian sandstones of the Shadrovskaya and Sedoyakhskaya formations are distinguished by the presence of a small admixture of hematite.

RESULTS OF PALEOMAGNETIC STUDY

The magnitude and direction of the vector of natural remanent magnetization (NRM) during the stepwise demagnetization in the alternating magnetic field (AF demagnetization) or thermal demagnetization (T-demagnetization) were measured in the 755 SRM cryogenic magnetometer (superconducting rock magnetometer) (2G Enterprises, United States) with a built-in unit for AF demagnetization. The T-demagnetization was conducted in the MMTD-80 shielded thermal demagnetizer (Magnetic Measurements, Britain). All the instruments are located in the shielded room with a residual field of ~250 nT in the Laboratory of geodynamics and paleomagnetism of the Trofimuk Institute of Petroleum Geology and Geophysics, Siberian Branch, Russian Academy of Sciences, Novosibirsk. The principal component analysis of NRM and the interpretation of the obtained results were conducted with the use of specialized software (Enkin, 1994; Torsvik and Smethurst, 1999) which use the known mathematical algorithms (Kirschvink, 1980; etc).

Since in all the studied rocks it was established that magnetic minerals undergo irreversible changes (crystallizations of the new magnetic phases) under heating, we tried, as far as possible, to use AF-demagnetization for identifying the NRM components. However, the up-to-date instruments are capable of creating the magnetic field up to at most 150 mT, which is not always sufficient for complete demagnetization of the samples due to the presence of grains with high magnetic hardness. Therefore, in many cases, despite the mineralogical changes on heating, we had to use the T-demagnetization.

Sandstones of the Shadrovskaya and Sedoyakhskaya formations. AF-demagnetization proved to be only efficient for the samples from the 14NZ01 outcrop (Fig. 2a). The almost complete removal of viscous magnetization takes place at the magnetic field of 5 mT. The median destructive field (MDF) is about 20 mT. The regular highly coercive component corresponds to the characteristic (ChRM) magnetization and is reliably diagnosed in the interval from 12 to 50 mT (sample 14nz007b in Fig. 5a). In stratigraphic coordinates, the established ChRM has a positive inclination of ~45° and eastern declination (Table 1).

The results of T-demagnetization of sandstones from the 14NZ02 and 14NZ03 outcrops (samples 14nz016a and 14nz033a in Fig. 5a, respectively) suggest that viscous magnetization is destroyed at ~150°C. Complete demagnetization (at least 90% of the initial NRM) is typically achieved on heating to above 580°C. The regular component, which as a rule corresponds to the characteristic one, is removed in the interval from 300 to 580°C and does not differ from the ChRM established in sandstones from the 14NZ01 sampling point based on the results of AF-demagnetization (Table 1). The precision of the site-mean ChRM direction distribution in ancient (stratigraphic) coordinates is significantly higher; however, $K_s/K_g = 4.27$ is below the critical value of 6.39 for the sample of three vectors according to (McElhinny, 1964). The correlation fold test (Enkin, 2003) is indeterminate due to the small difference in the bedding elements of the stratum. Nevertheless, the simulation parametric fold test (Watson and Enkin, 1993) shows that the maximal precision is achieved at 99.6% unfolding (Fig. 5b) but the 95% confidence interval with the maximal number of the simulations (1000) is 38.7%. In these cases, when the bedding elements of the stratum differ insignificantly, the most reliable result can be yielded by

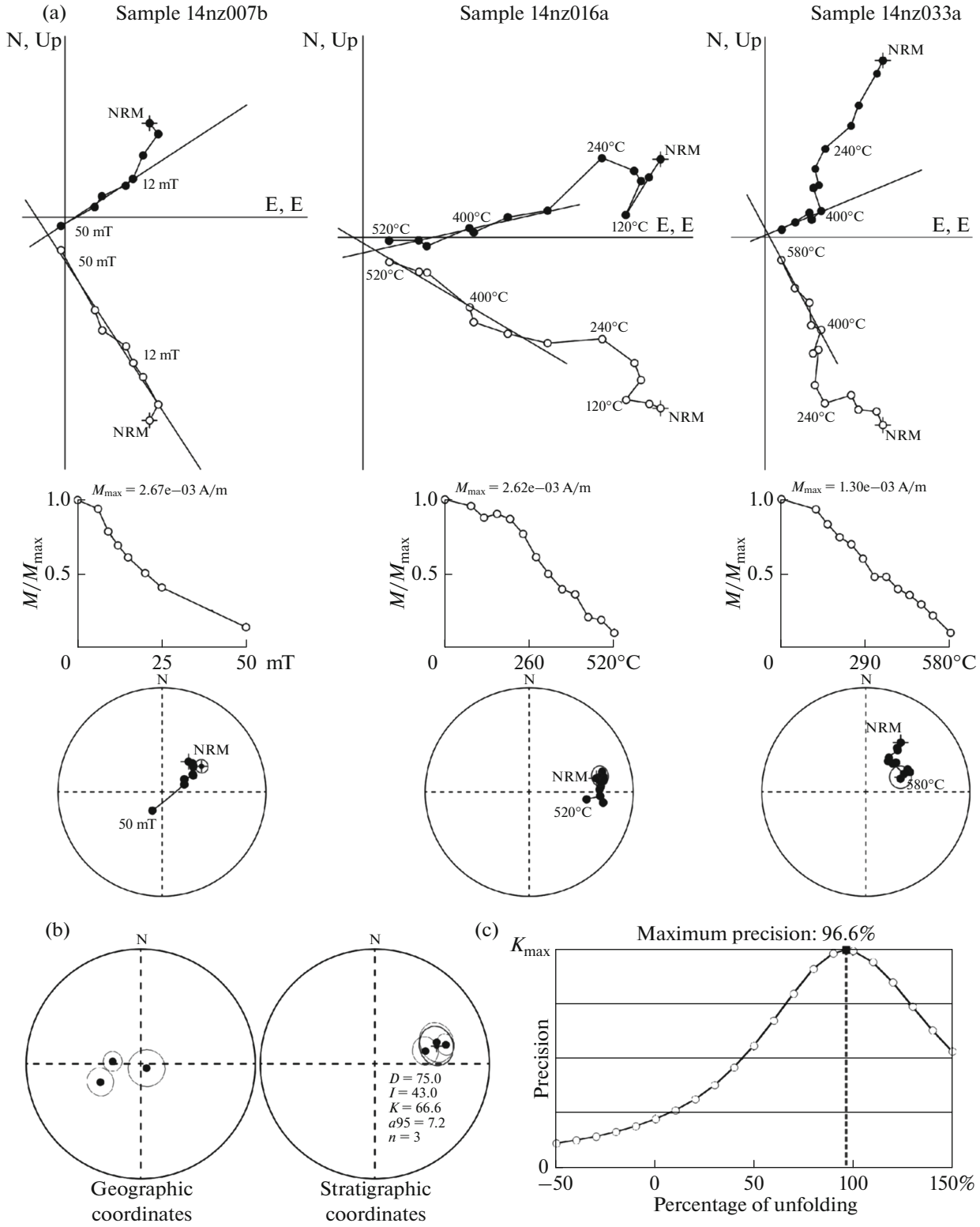


Fig. 5. Results of component analysis for sandstones of Shadrovskaya and Sedoyakhskaya formations: (a) characteristic orthogonal diagrams (hereinafter, gray lines indicate direction of stable component, stratigraphic coordinates), corresponding graphs of NRM variations during demagnetization, and stereograms (hereinafter, direction of established component is shown in gray); (b) distribution of site-mean directions; (c) graph of variations in precision parameter under unfolding (result of fold test in modification (Watson and Enkin, 1993)).

Table 1. Paleomagnetic directions for Paleozoic sediments of Novaya Zemlya Archipelago

Formation, sampling point name, and coordinates	*n*/*N*	Geographic coordinates		Stratigraphic coordinates		*k*	α_{95}
		D, °	*I*, °	*D*, °	*I*, °		
Shadrovskaya Formation, 14NZ01 72.05° N, 51.56° E	5/12	245.8	57.8	76.2	52.0	63.2	9.7
Shadrovskaya Formation, 14NZ02 72.05° N, 51.56° E	10/13	270.0	70.2	77.4	35.5	68.0	5.9
Sedoyakhskaya Formation, 14NZ03 72.06° N, 51.57° E	11/11	129.6	84.9	71.4	41.6	13.1	13.1
Mean over Shadrovskaya and Sedoyakhskaya formations	(3)	250.4	73.7			15.6	16.4
				75.0	43.0	66.6	7.2
Barentsevskaya Formation, 14NZ07 76.26° N, 61.78° E	8/10	149.1	35.3	54.3	51.7	45.5	8.3
Barentsevskaya Formation, 14NZ08 76.26° N; 61.78° E	9/12	330.5	−38.3	229.2	−51.4	205.4	3.6
*Barentsevskaya Formation Mean	(2)	149.8	36.9	51.5	51.6	80.1	4.0
Konglomeratovaya Formation, 14NZ05, 76.21° N, 62.92° E	11/18	296.6	41.1	308.8	−8.0	39.0	7.4
Ermolaevskaya Formation, 14NZ10, 76.18° N, 62.81° E	10/11	139.0	68.1	142.5	17.2	259.3	3.0

n/*N* is ratio of number of samples used in statistics to total number of studied samples (sites for calculating site mean); *D* is declination; *I* is inclination; *k* is precision parameter; α_{95} is 95%-confidence oval; * after reduction to common polarity.

the correlation test in the modification (McFadden, 1990). The critical value of the correlation coefficient ξ_2 at the 95% confidence level for $n = 3$ is 2.076, ξ_2 in geographical coordinates (2.883) is above the critical value and in the stratigraphic coordinates (0.167) it is below the critical value, which suggests the prefolding age of the magnetization.

Barentsevskaya Formation. The component composition of NRM in the sandstones from the 14NZ07 outcrop (Fig. 2b) is studied based on the results of AF-demagnetization (sample 14nz084b in Fig. 6a). The studied samples are characterized by a concave shape of the magnetization decay curves reflecting the rapid reduction in NRM under the action of an alternating magnetic field with a relatively low amplitude. The MDF value is about 15 mT. The direction of the stable component can be detected even starting from this point; however, the most straight linear segments directed to the origin of the coordinates are visible in the orthogonal diagram starting from ~40 mT. The revealed ChRM vector has a regular orientation with a relatively steep positive inclination and differs by the northeastern declination from the ChRM direction in the sandstones of the Shadrovskaya and Sedoyakhskaya formations (Table 1).

Considering the high magnetic hardness of the sandstones from the 14NZ08 outcrop, we used T-demagnetization for identifying their magnetization components. The NRM of these samples is fairly stable to heating. Frequently, heating up to 610–670°C only removes at most half of the initial NRM value (sample 14nz092a in Fig. 6a). Here, the remanent magnetization vector in the highest temperature (hematite) interval has a chaotic orientation, whereas the regular component can only be isolated on the linear intervals typically characterizing the interval from 400 to 610°C. In contrast to the ChRM of the sandstones sampled at the 14NZ07 point, the isolated component has a negative southwestern declination (Table 1, Fig. 6b). The angle between the mean directions of the inclination having different signs after the reversal is $\gamma = 3.3°$ with the critical angle $\gamma_c = 8.2°$, which corresponds to class B in the classification (McFadden and McElhinny, 1990). This indicates that the rocks acquired the identified component during at least two magnetic polarity epochs and are unlikely to contain superimposed magnetization, i.e., the time of the fixation of the paleomagnetic record corresponds to the geological age of the Barents Formation.

Konglomeratovaya Formation. All the samples were subjected to T-demagnetization, the results of which indicate that the NRM has a complicated component composition (Fig. 7a). Complete demagnetization is typically achieved on heating to 580°C. The medium-temperature component is isolated in the temperature interval from 320 to 400°C and has a chaotic distribu-

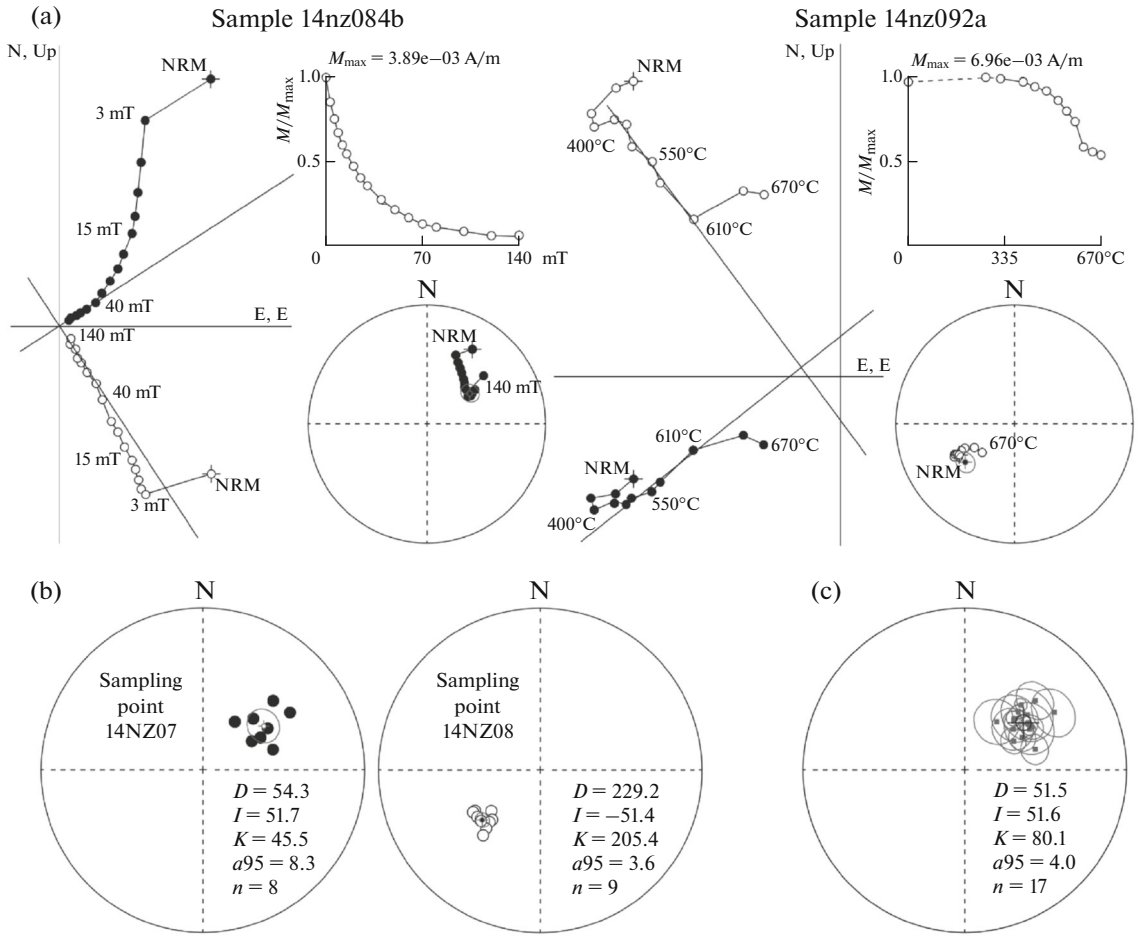


Fig. 6. Results of component analysis for sandstones of Barentsevskaya Formation: (a) characteristic orthogonal diagrams, corresponding graphs of NRM variations during demagnetization, and stereograms; (b) distribution of established magnetization component by paleomagnetic sampling sites; (c) distribution of all unit vectors after reducing them to common polarity.

tion of the magnetization directions. The high-temperature component corresponds to the characteristic one, is removed in the temperature interval from 400 to 580°C, and has a small (by magnitude) regular negative inclination and northwestward declination (Table 1).

Ermolaevskaya Formation. Besides the viscous component which is destroyed below 200°C, the NRM of these sandstones contains only one stable component. From 200°C up to the complete demagnetization at ~600°C, the magnetization gradually decreases; its direction barely changes and corresponds to the characteristic magnetization direction (Fig. 7b). These main features in the behavior of the NRM vector during T-demagnetization are repeated for almost all the samples.

DISCUSSION

The paleomagnetic reliability of the directions of the stable magnetization components revealed by our analysis for the different studied regions of the archipelago significantly differs. The determination from the sandstones of the Shadrovskaya and Sedoyakhskaya formations is perhaps most reliable (Table 1, Fig. 5). Firstly, the presence of an identical stable component in this case is most soundly substantiated from the statistical standpoint. Secondly, the revealed component is present in the rocks that characterize quite a large interval of the section and were formed during at least a few million years, which means that the secular variations are averaged out. Thirdly, the magnetic properties of the rocks testify to the detrital nature of the magnetization. Finally, the primary origin of the revealed magnetization is supported by such an important argument as the positive fold test. Compared to the previous determinations for which only an indeterminate result of the reversal test was obtained, the new data better substantiate the position of the paleomagnetic pole for the Southern Novaya Zemlya tectonic segment of the Paikhoi–Novaya Zemlya region (Table 2). Considering the data about the age of the magnetization for all the existing determinations,

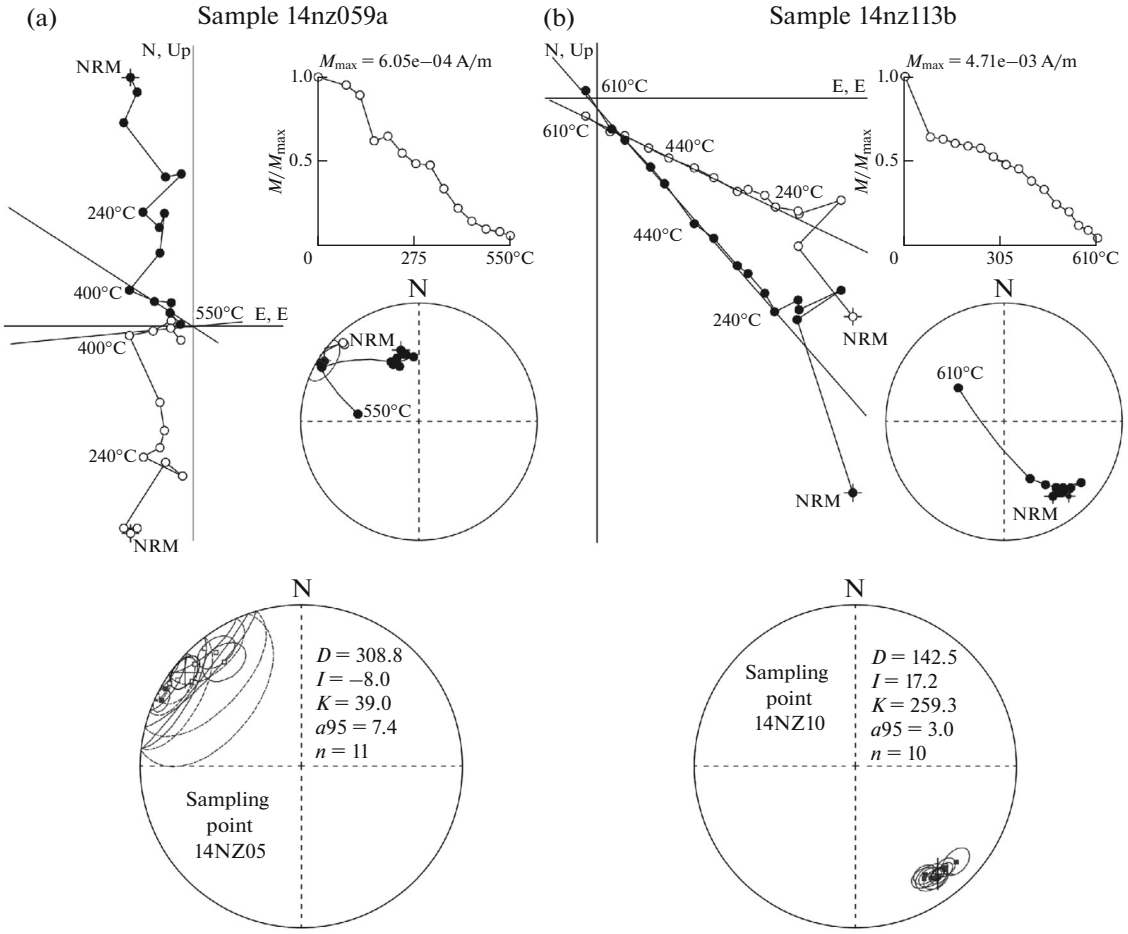


Fig. 7. Representative orthogonal diagrams, graphs of NRM variations during demagnetization, and stereograms based on results of demagnetization; bottom: distribution of established component over paleomagnetic sampling points for (a) Konglomeratovaya Formation and (b) Ermolaevskaya Formation.

we assume that the mean pole characterizes the boundary of 250 Ma.

The coordinates of the paleomagnetic poles for the same time interval in the case of the Northern Novaya Zemlya segment (Gurevich and Slautsitais, 1984; *Paleomagnitnye*..., 1986) are, within the limits of the determination error, the same as the coordinates of the Southern Novaya Zemlya poles. However, these data are not particularly reliable (Table 2) and are therefore disregarded in the subsequent tectonic analysis.

A fairly reliable paleomagnetic result is also obtained from the Late Carboniferous sandstones of the Barentsevskaya Formation. According to the geological data, the studied interval of the section corresponds to the beginning of the Kiama superchron; nevertheless, at point 14NZ07 we see the normal polarity (Table 1, Fig. 6). Here, the positive reversal test testifies to the primary origin of the magnetization. Besides, the presence of the magnetization directions of different polarity in the close outcrops indicates that the secular variations are likely to be averaged out. The age of the rocks and the age of the magnetization revealed in these rocks remain unclear. On one hand, the revealed normal polarity can indicate a somewhat older interval of fixation of the paleomagnetic signal of about 320 Ma ago before the beginning of Kiama. On the other hand, the evolution of the Earth's magnetic field in the Late Carboniferous has been poorly studied and the magnetostratigraphy of this stage of the geological evolution has been debatable up to now. In a number of publications, it is shown that there were several short episodes of normal polarity at the beginning of Kiama. One of these episodes is located in the immediate vicinity of the Carboniferous/Permian boundary, i.e., ~300 Ma ago (Menning et al., 1988; Opdyke and Channell, 1996; Alva-Valdiviaa et al., 2002). In this case, the estimates for the age and magnetization of the rocks suggested by us are valid and the section of the Barents Formation itself can be very important for studying the evolution of the geomagnetic field at the beginning of one of the largest chrons of normal polarity. Without downgrading the potential significance of this determination, we note that due to the very limited statistics this result

Table 2. Paleomagnetic poles for Upper Paleozoic rocks of Novaya Zemlya Archipelago

No.	Object	*n*	Test	Paleomagnetic pole				Displacement		Source
				Plat	*Plon*	A_{95}	*T*	*F*	*R*	
Southern Novaya Zemlya segment										
1	Sandstones of Shadrovskaya and Sedoyakhskaya formations	3/23	F+	28.3	147.5	7.0	260–250			(This work)
2	Sandy clay deposits #4612	1/44	Ro	32	153	6.0	251–245			(*Paleomagnitnye*..., 1986)
3	Sandy clay deposits #4610	1/47	Ro	19	149	8.0	250–245			(*Paleomagnitnye*..., 1986)
Novaya Zemlya mean (1–3) pole				26.5	149.8	10.9	250	23.4 ± 8.4	14.8 ± 9.5	(This work)
Baltica mean pole				52.8	152.8	3.7	250			(Torsvik et al., 2012)
4	Gabbro-dolerites #5193	2/26	Ro	–8	139	10.0	385–359	8.2± 10.4	11.4 ± 10.4	(Pogarskaya, 1984)
Baltica mean pole				3.3	147.3	10.0	370			(Torsvik et al., 2012)
Northern Novaya Zemlya segment										
5	Sandy clay deposits #4841	3/32		21	161	9.3	263–251			(*Paleomagnitnye*..., 1986)
6	Sandy clay deposits #4613	1/81	Ro	31	146	7.0	251–245			(*Paleomagnitnye*..., 1986)
7	Sandstones of Barents Formation	2/17	Rb	40.2	182.0	4.5	310–300	4.1 ± 3.8	12.0 ± 4.4	(This work)
Baltica mean pole				41.5	167.9	2.3	300			(Torsvik et al., 2012)
8	Sedimentary rocks and gabbro-diabases #5222	5/87	F+, M, Ro	–3.8	126.5	13	423–359			(Pogarskaya and Iosifidi, 1986)
9	Sandstones of Konglomeratovaya Formation	1/11		–4.6	114.2	5.3	388–372			(This work)
10	Sandstones of Ermolaevskaya Formation	1/10		–2.2	99.8	2.2	419–407			(This work)
Novaya Zemlya mean (8–10) pole				–3.6	113.5	20.6	390	–6.8 ± 17.1	28.7 ± 17.2	(This work)
Baltica mean pole				–4.0	143.0	11.0	390			(Torsvik et al., 2012)

No. corresponds to pole numbers in Fig. 7; # is number of determination in catalog (*Paleomagnitnye*..., 1986); *n* is number of objects/samples used in determining pole; test: *R* is reversal test (index o denotes indeterminate result, index b denotes class B according to (McFadden and McElhinny, 1990), F+ stands for positive fold test; M magnetic mineralogy is substantiated; paleomagnetic pole: *Plat, Plon* are geographical latitude (degree North) and longitude of pole (degree East), A_{95} is radius of 95% confidence circle, *T* is assumed age of determination, Ma. Displacement denotes quantitative parameters of displacement relative to Baltica craton), in degrees: *R* is angle of clockwise (+) and counterclockwise (–) rotation, *F* describes latitudinal displacement (along longitude towards pole) in northward (+) and southward (–) direction. Calculations are conducted with use of algorithm that is implemented in PMSGC v.4.1 (Enkin, 1994).

can only be considered as preliminary and requiring validation.

Finally, the least reliable determinations are obtained for the Devonian interval; they can only be considered as very approximate. The inclination of the magnetization vector of the stable components in the sandstones of the Ermolaevskaya and Konglomeratovaya formations in the stratigraphic coordinates has different signs (Table 1); i.e., these rocks were formed during the epochs with a different polarity of the geomagnetic field. However, the angle between the mean directions after the reduction to the common polarity $\gamma = 16.2$ is twice as high as the critical angle $\gamma_c = 8.0$ according to (MaFadden and McElhinny, 1990). Here, the negative reversal test does not preclude the primary origin of the magnetization of both formations and can be associated with a significant difference in the age of the rocks. The ancient genesis of the revealed components can be supported by the results of the rocks' magnetic study and the closeness of the coordinates of the virtual geomagnetic poles calculated from the directions of the revealed stable NRM components in the ancient coordinate system with the determination for the Devonian of the Northern Novaya Zemlya segment (Table 2) (Pogarskaya and Iosifidi, 1986). These data are obtained from the set of intrusive and sedimentary rocks and are perhaps most soundly substantiated from the standpoint of paleomagnetic reliability. Here, the primary nature of the magnetization is demonstrated by the mineralogical evidence and the fold test is positive. A disadvantage of this determination lies in the wide age interval of the rocks and related magnetization which is probably responsible for the indeterminate reversal test and fairly large confidence oval (Table 2). Nevertheless, the coordinates of the newly obtained and previous Devonian poles of the Northern Novaya Zemlya segment differ insignificantly. For the tectonic interpretation of these data, we averaged these coordinates and assumed the age of the mean geomagnetic pole at 390 Ma (Table 2).

For the Devonian of the Southern Novaya Zemlya segment, there is only one paleomagnetic determination as of now (Pogarskaya, 1984). This determination is not particularly reliable and its applicability in the tectonic analysis is problematic. However, we use these data for an approximate estimate due to the absence of other information and assume the age of the virtual geomagnetic pole to be 370 Ma (*Paleomagnitnye*..., 1986).

The present curved configuration of the Pai-Khoi–Novaya Zemlya fold-nappe area is definitely secondary. The collision of the Svalbard Plate with the continental margin of Baltica and the formation of the Cadomian accretionary–collisional edifices of the Bol'shezemel'skaya part of the Timan–Pechora basin's basement west of the Baidaratskii tectonic suture occurred at the Precambrian/Paleozoic boundary. Despite the difference in their basements, the Paleozoic sedimentary sections, the Southern Novaya Zemlya segment, and Northern Novaya Zemlya segment of the region were formed in the same tectonic conditions—the shelf of the Uralian Ocean (Metelkin et al., 2015). Hence, the Paleozoic geological complexes of the Pai-Khoi–Novaya Zemlya area are an analog of the West Uralian fold zone the tectonic elements of which are oriented submeridionally. The deformations that resulted in the transformation of the lineament pattern into a curved horseshoe structure could be caused by the presence of a rigid block with a characteristic shape, which was embedded into the structure of the region at the time of a collisional or postcollisional event in the Triassic (Shatsillo, 2015), or by the combination of the compression and extension regimes emerging against the background shear in the intraplate tectonic conditions. Since the main events related to the collision within the Uralian orogen are not younger than to the Permian–Early Triassic (Puchkov, 2003; Samygin and Burtman, 2009), the Early Cimmerian and later deformations in the Pai-Khoi–Novaya Zemlya region were most likely to be due to the postcollisional intraplate shear setting. In our opinion, the latter could have resulted from the transformation of Central Asia's structure in the conditions of a sinistral shear of the Kazakhstan and Siberian tectonic domains relative to the European domain under a common clockwise rotation of the Eurasian Plate, as it is assumed in the model (Meytelkin et al., 2010). This shear system probably has a direct manifestation in the structure of the Pai-Khoi–Novaya Zemlya region—the Baidaratskii fault separating its Southern Novaya Zemlya and Northern Novaya Zemlya segments. The shears could have caused or accompanied the opening of the Mesozoic rift system in West Siberia and initiated the formation of the South Kara basin where the extension associated with these shears reached its maximum. The simultaneous growth of the curved foldand-thrust structure of the Novaya Zemlya uplift in this case could be due to the associating transpression.

We suggest verifying the discussed tectonic model by the existing paleomagnetic data (Table 2). To this end, we compared the existing determinations for the Novaya Zemlya Archipelago with the apparent polar wander path (APWP) of the geomagnetic pole of Baltica (Fig. 8). First of all we can see the different pattern of mutual relations of the poles of the Southern Novaya Zemlya and Northern Novaya Zemlya segments with the European APWP, which is fairly natural in the context of the discussed model.

The locations of the poles of the Northern Novaya Zemlya fragment insignificantly differ from the corresponding poles of Baltica. The differences are systematic and only imply clockwise rotation of the Northern Novaya Zemlya segment relative to the craton by an angle of 10°–15° (Table 2). Parameter R, describing the amount of the rotation of the Northern Novaya

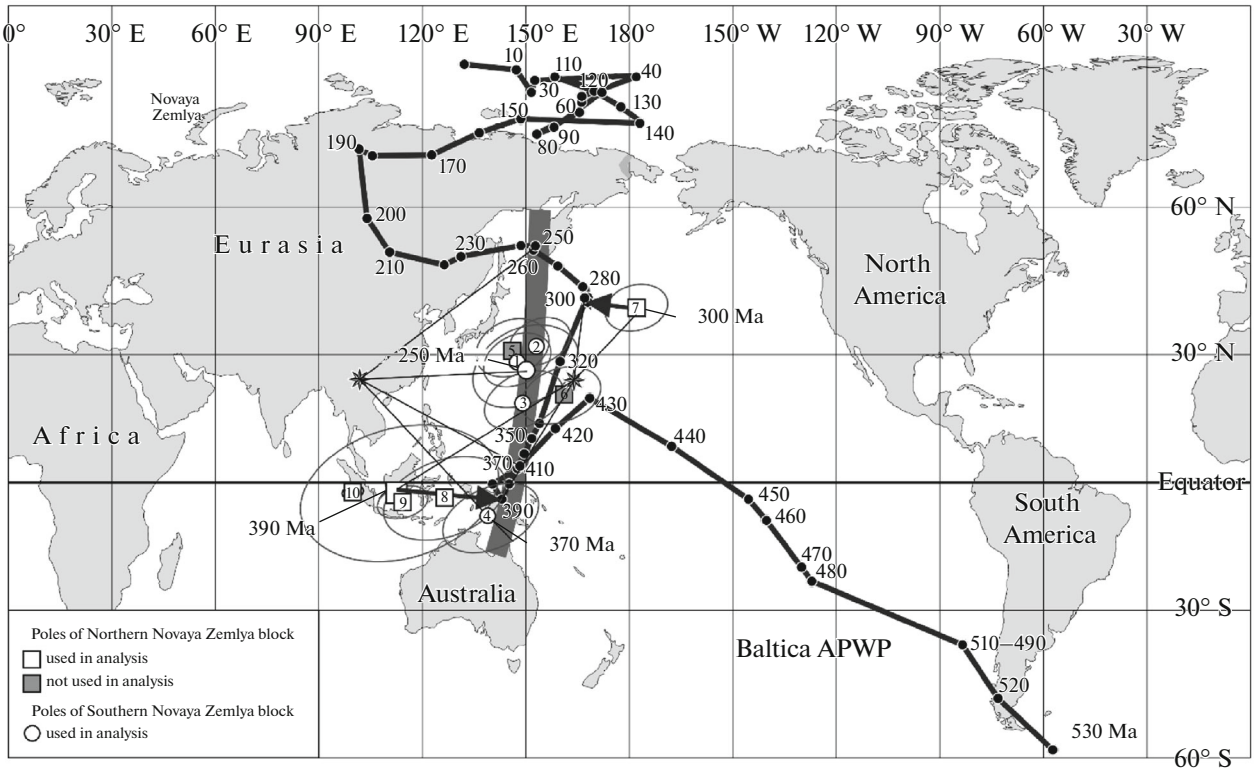


Fig. 8. Position of Late Paleozoic paleomagnetic poles for territory of Novaya Zemlya Archipelago compared to APWP for Baltica (Torsvik et al., 2012). Numerals near poles indicate age in Ma. Pole numbers correspond to numbering in Table 2. Explanations are presented in text.

Zemlya segment relative to the craton, in the case of the comparison of the Devonian poles, is somewhat larger, by 300 Ma compared to the Late Carboniferous (Table 2). However, this is most likely due to the less accurate determination of the age and coordinates of the pole itself. In the suggested model, the revealed rotation can be tectonically accounted for by the frontal compression in the submeridional direction (northwestward in the present (geodetic) coordinates) oriented along the Baidaratskii shear.

The kinematics of the relative displacement of the Southern Novaya Zemlya segment is different. Besides the similar or slightly smaller rotation, there is also a significant latitudinal (along the longitude) northward displacement which is reflected in parameter F (Table 2). For the most reliable 250-Ma pole, F is 23.4° ± 8.5°. With allowance for the determination error, this F value implies the displacement of the Southern Novaya Zemlya segment by more than 1500 km, which is barely realistic. Although the relative positions of the Devonian poles of the Southern Novaya Zemlya segment and Baltica are similar to the Late Permian, in fact they all lie on the great-circle arc (Fig. 8), and displacement F at the 370-Ma boundary is within the limits of the determination error (Table 2). Therefore it can be hypothesized that the F value is largely determined by the effect of inclination shallowing (Kodama, 2012). Unfortunately, the scarcity of the statistics for this determination prevents us from estimating the contribution of this effect based on the present techniques (Tauxe et al., 2008). On the other hand, considering the low reliability of the Devonian pole and the sufficiently well substantiated result for the Permian/Triassic boundary, we have grounds to believe that the inclination shallowing only partly accounts for the established discrepancy of the poles of the Southern Novaya Zemlya and Europe. A certain (probably insignificant, a few hundred km) fraction of F could correspond to the real displacement at the time of the deformation. This estimate primarily follows from the general regional tectonic model underlying the interpretation (Metelkin et al., 2010). The key role determining the style of the deformations can be played by the predominant transpression regime emerging within the Southern Novaya Zemlya segment under shear displacements along the Baidaratskii fault. This lateral compression resulted in splitting the structure of the Southern Novaya Zemlya segment into separate tectonic slices which are not only thrust over each other in the sublatitudinal (in the present coordinates) direction but also submeridionally shifted according to the main displacement direction along the Baidaratskii fault.

CONCLUSIONS

Our rock magnetic and paleomagnetic studies have shown that the Upper Paleozoic sediments of the Novaya Zemlya Archipelago are a potentially promising object for magnetotectonic analysis and unraveling the geological history of the Barents–Kara region of the Arctic. However, the tectonic interpretation of the paleomagnetic data should tale into account the probable inclination shallowing for some local structures and stratigraphic units.

The new paleomagnetic determinations obtained by our analysis support the results of the previous studies. The coordinates of the virtual geomagnetic poles for the interval of 250–260 Ma in the Southern Novaya Zemlya segment and 410–390 Ma in the Northern Novaya Zemlya segment of the Pai-Khoi–Novaya Zemlya fold-and-thrust area fairly well agree with the previously calculated coordinates. For the first time, paleomagnetic substantiation is obtained for the boundary of 310–300 Ma for the Northern Novaya Zemlya segment. Here, the studied section of the Barents Formation contains signs that can substantiate a narrow zone of normal polarity near the Carboniferous/Permian boundary, which is potentially important for solving the problems of regional magnetostratigraphy and interpreting the regularities in the evolution of the Earth's magnetic field at the stage of its stabilization at the beginning of the long stationary Kiama chron of reversed polarity.

Based on the entire set of the existing data, a significant difference is revealed in the kinematics of the displacements and, hence, in the deformation mechanisms of the Southern and Northern Novaya Zemlya segments of the Pai-Khoi–Novaya Zemlya region. The Early Cimmerian deformation stage was probably caused by the sinistral strike-slip displacement along the Baidaratskii fault during which the internal structure of the Southern Novaya Zemlya segment could experience both the nappe-thrust and shear deformations. The Northern Novaya Zemlya segment, which shifted relative to the Southern one, was mainly deformed in the thrusting mode with a general clockwise rotation of this segment with respect to the East European craton.

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